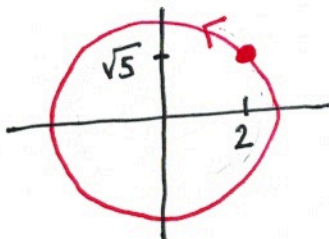


Section 3.9 Related Rates

Ex ① An object travels around a circle of radius 3. Find $\frac{dx}{dt}$ if $\frac{dy}{dt} = 2 \frac{\text{ft}}{\text{sec}}$

When $x = 2$.



$$x^2 + y^2 = 9$$

$$2^2 + y^2 = 9$$

$$y^2 = 5$$

$$y = \sqrt{5}$$

Take derivative...

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

$$2(2) \frac{dx}{dt} + 2\sqrt{5} \cdot 2 = 0$$

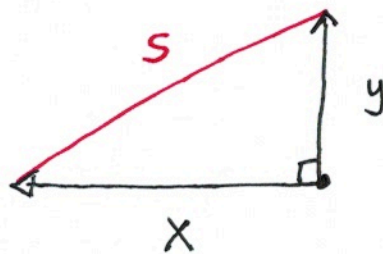
$$4 \cdot \frac{dx}{dt} = -4\sqrt{5}$$

$$\frac{dx}{dt} = -\sqrt{5} \text{ or } -2.236 \frac{\text{ft}}{\text{sec.}}$$

Ex ② Two runners start from the same place.

One runs North at 8 mph while the other runs west at 10 mph.

How fast is the distance between them increasing after 2 hours? Find $\frac{ds}{dt}$



$$t = 2$$

$$x = 20 \text{ mi.}$$

$$y = 12 \text{ mi.}$$

$$20^2 + 12^2 = s^2$$

$$23.324 = s_{\text{mi.}}$$

$$x^2 + y^2 = s^2$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 2s \cdot \frac{ds}{dt}$$

$$2(20) \cdot 10 + 2(12) \cdot 8 = 2(23.324) \cdot \frac{ds}{dt}$$

$$400 + 192 = 2(23.324) \cdot \frac{ds}{dt}$$

$$\frac{592}{2(23.324)} = \frac{ds}{dt}$$

$$(12.691 \text{ mph})$$

Ex ③ A basketball is being inflated at a rate of $30 \frac{\text{in}^3}{\text{sec}}$. How fast is the radius increasing when $r = 4.7 \text{ in}$?

Find $\frac{dr}{dt}$

$$V = \frac{4}{3} \pi r^3$$

$$\frac{dV}{dt} = \frac{4}{3} \pi \cdot 3r^2 \cdot \frac{dr}{dt}$$

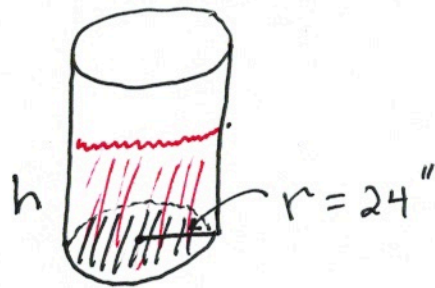
$$30 = \frac{4}{\cancel{3}} \pi \cdot \cancel{3} (4.7)^2 \cdot \frac{dr}{dt}$$

$$\frac{30}{4\pi(4.7)^2} = \frac{dr}{dt}$$

$$0.108 \frac{\text{in}}{\text{sec}} = \frac{dr}{dt}$$

Ex ④ A right circular cylinder is being filled with water at a rate of $210 \frac{\text{in}^3}{\text{sec}}$.

How fast is the water level rising if the radius is 2 feet?
(24 inches)



$$V = \pi \cdot r^2 \cdot h$$

$$V = \pi (24)^2 \cdot h$$

$$\frac{dV}{dt} = \pi (24)^2 \cdot \frac{dh}{dt}$$

$$210 = \pi (24)^2 \frac{dh}{dt}$$

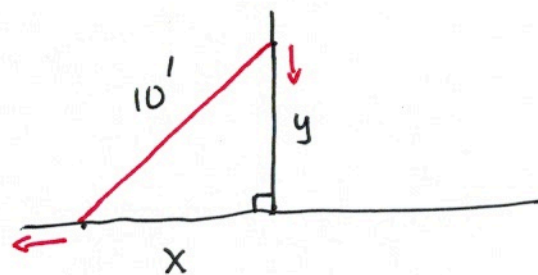
$$\frac{210}{\pi (24)^2} = \frac{dh}{dt}$$

$$0.116 \frac{\text{in}}{\text{sec}} = \frac{dh}{dt}$$

Ex ⑤ A 10 foot pole is leaning against a wall. The top of the pole slides down the wall at $3 \frac{\text{in.}}{\text{sec.}}$. How fast is the bottom sliding out when it is 7 feet from the wall?

$$\frac{3}{12} = \frac{1}{4}$$

$$\frac{3 \text{ in}}{\text{sec}} = \frac{1}{4} \frac{\text{ft}}{\text{sec}}$$



$$x = 7$$

$$7^2 + y^2 = 100$$

$$y^2 = 51$$

$$y = \sqrt{51}$$

$$x^2 + y^2 = 100$$

$$2x \cdot \frac{dx}{dt} + 2y \cdot \frac{dy}{dt} = 0$$

$$2x \cdot \frac{dx}{dt} = -2y \cdot \frac{dy}{dt}$$

$$\frac{dx}{dt} = - \frac{y \cdot \frac{dy}{dt}}{x}$$

$$\frac{dx}{dt} = - \frac{\sqrt{51} \cdot (-3)}{7}$$

$$\frac{dx}{dt} = 3.061 \frac{\text{ft}}{\text{sec}}$$